

Factorizations in Number Theory

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1 Relations of type $ab = cd$ and applications

Lemma. If a, b, c, d are positive integers such that $ab = cd$, then there exist positive integers m, n, p, q such that $(n, p) = 1$ and

$$a = mn, b = pq, c = mp, d = nq.$$

Proof. We have

$$ab = cd \implies \frac{a}{c} = \frac{d}{b} = \frac{n}{p}, \quad \text{where } (n, p) = 1.$$

Now $n \mid (a, d)$ and $p \mid (b, c)$ hence the choice $m = \frac{a}{n} \in \mathbb{N}^*$, $q = \frac{b}{p} \in \mathbb{N}^*$ works. $\square$

This simple but useful lemma essentially gives us a *parametrization* for all solutions of the equation $ab = cd$. Note that the techniques employed here apply to any unique factorization domain.

Example 1. Let $a, b, c, d \in \mathbb{N}^*$ be such that $ab = cd$. Then $a + b + c + d$ is composite.

Solution. By the lemma, there exist $m, n, p, q \in \mathbb{N}^*$ such that $a = mn, b = pq, c = mp, d = nq$. Then $a + b + c + d = (m + q)(n + p)$, which is composite. $\square$

Example 2 (Pythagorean Triples). Find all triples $x, y, z \in \mathbb{N}^*$ such that

$$x^2 + y^2 = z^2$$

Solution. We assume $(x, y) = 1$, so that $(y, z) = (z, x) = 1$. Rewrite the equation as

$$x^2 = (z - y)(z + y)$$

By the lemma, there exist $m, n, p, q \in \mathbb{N}^*$ such that $x = mn = pq, z + y = mp, z - y = nq$ and $(n, p) = 1$. Now $mn = pq \implies n \mid pq \xrightarrow{(n, p)=1} n \mid q$ and therefore $q = nk, m = pk$ for some $k \in \mathbb{N}^*$. Hence

$$z = \frac{mp + nq}{2} = k \frac{p^2 + n^2}{2}, \quad y = k \frac{mp - nq}{2} = k \frac{p^2 - n^2}{2}, \quad x = knp$$

For $k = 2k_1$ even, we obtain the solution family

$$z = k_1(p^2 + n^2), \quad y = k_1(p^2 - n^2), \quad x = 2k_1np \quad \text{where } (n, p) = 1$$

For k odd, n and p have the same parity and $u = \frac{n+p}{2}, v = \frac{p-n}{2}$ are coprime. Hence we obtain

$$z = k \frac{p^2 + n^2}{2} = k(u^2 + v^2), \quad y = k \frac{p^2 - n^2}{2} = 2kuv, \quad x = k(u^2 - v^2)$$

– however these solutions have already been listed above. $\square$

Example 3. Show that if $a > c > d > b > 0$ are integers such that $a^2 + b^2 = c^2 + d^2 = x$, then x is composite.

Solution. Rewrite the condition $a^2 + b^2 = c^2 + d^2$ as

$$a^2 - c^2 = d^2 - b^2 \implies (a - c)(a + c) = (d - b)(d + b).$$

By the lemma, there exist m, n, p, q such that $(n, p) = 1$ and

$$a + c = mn, \quad a - c = pq, \quad d + b = mp, \quad d - b = nq$$

and note that $a > c > d > b$ implies $n > p, m > q$. Therefore,

$$a = \frac{mn + pq}{2}, \quad b = \frac{nq - mp}{2} \implies x = a^2 + b^2 = \frac{(m^2 + q^2)(n^2 + p^2)}{4}.$$

Since $m^2 + q^2$ and $n^2 + p^2$ are at least $2^2 + 1^2 = 5$, one can easily see that x cannot be prime. $\square$

Example 4 (IMO 2001). Let $a > b > c > d > 0$ be integers such that

$$a^2 + c^2 - ac = b^2 + d^2 + bd$$

Show that the integer $ab + cd$ is composite.

Solution. Rewrite the given relation as

$$(a + c)^2 + 3(a - c)^2 = (b - d)^2 + 3(b + d)^2 \implies$$

$$(a + c + b - d)(a + c - b + d) = 3(a - c + b + d)(b + d - a + c)$$

We assume $3 \mid a + c + b - d$ (the case $3 \mid a + c - b + d$ is analogous). Then there exist $m, n, p, q \neq 0$ with $(n, p) = 1$ such that

$$a + c + b - d = 3mn, \quad a + c - b + d = pq, \quad a - c + b + d = mp, \quad b + d - a + c = nq$$

Note that

$$b + d - a + c < a + c - b + d < a - c + b + d < a + c + b - d \implies n < p < 3n \text{ and } m > q \quad (1)$$

Solving for a, b, c, d , we obtain

$$\begin{aligned} a &= \frac{3mn + pq + mp - nq}{4} & b &= \frac{mp + nq + 3mn - pq}{4} \\ c &= \frac{3mn + nq + pq - mp}{4} & d &= \frac{mp + nq + pq - 3mn}{2} \end{aligned}$$

and therefore

$$ab + cd = \frac{(3m^2 + q^2)np}{4}$$

Assume by contradiction that $ab + cd$ is not composite. We distinguish 2 possibilities:

1) $m + q$ is even. Then $4 \mid 3m^2 + q^2$, and hence either $3m^2 + q^2 = 4$ or $n = p = 1$, both impossible by (1).

2) $m + q$ is odd. Then $3m^2 + q^2$ has to be prime and $np = 4$ which is again impossible since $n < p < 3n$ by (1). $\square$

Example 5. Show that the following system has no solutions in positive integers:

$$\begin{cases} x^2 + y^2 = u^2 & (1) \\ x^2 - y^2 = v^2 & (2) \end{cases}$$

Solution. Assume (x, y, u, v) satisfies the system, and choose one with $x + y$ minimal. Note that x, y, u, v are pairwise coprime. Considering mod 4, it follows that x is odd, y is even and u, v are odd. Recalling Pythagorean triples, we know that

$$x = c^2 - d^2, y = 2cd, x = a^2 + b^2, y = 2ab$$

for some $a, b, c, d > 0$ with $(a, b) = (c, d) = 1$. Now, since $ab = cd = \frac{y}{2}$, applying the lemma again we find the positive integers $m, n, p, q \in \mathbb{N}^*$ such that

$$a = mn, b = pq, c = mp, d = nq.$$

and m, n, p, q are pairwise coprime. Now,

$$a^2 + b^2 = c^2 - d^2 \implies m^2b^2 + p^2q^2 + n^2q^2 = m^2p^2 \implies m^2(n^2 - p^2) = q^2(p^2 + n^2)$$

Since $(n, p) = 1$, we have $(n^2 - p^2, n^2 + p^2) \in \{1, 2\}$. If $(n^2 - p^2, n^2 + p^2) = 1$, then $n^2 - p^2 = q^2, n^2 + p^2 = m^2$ and (n, p, m, q) is a solution of the original system, smaller than (x, y, u, v) , contradiction. Therefore, $(n^2 - p^2, n^2 + p^2) = 2$, n, p are odd and $\frac{n^2 - p^2}{2} = q^2, \frac{n^2 + p^2}{2} = m^2$ and now (m, q, n, p) is a smaller solution to the original system. $\square$

2 Working in Quadratic Field Extensions

Recall that the *algebraic integers* (i.e. the roots of monic polynomials with integer coefficients) inside a quadratic extension $\mathbb{Q}[\sqrt{d}]$ of the field $\mathbb{Q}$ form a ring. For $d \equiv 2, 3 \pmod{4}$ this ring is just $\mathbb{Z}[\sqrt{d}] = \{a + b\sqrt{d} \mid a, b \in \mathbb{Z}\}$. For $d \equiv 1 \pmod{4}$ this ring is $\mathbb{Z}[\frac{1+\sqrt{d}}{2}]$.

Some of these rings ($\mathbb{Z}[i] = \mathbb{Z}[\sqrt{-1}]$, $\mathbb{Z}[\sqrt{3}]$ and others) behave much like the integers: there is a multiplicative norm, Euclidean algorithm, greatest common divisor and most importantly, unique factorization. This enables one to employ most techniques valid for integers.

Theorem 1 (Baker-Heegner-Stark). *The ring of integers of $\mathbb{Q}[\sqrt{d}]$ with $d < 0$ is a unique factorization domain if and only if*

$$d \in \{-1, -2, -3, -7, -11, -19, -43, -67, -163\}.$$

For $d > 0$, it is not even known if there are infinitely many such rings (one reason why this is tricky is that there are infinitely many units). Here is a list of d for which $\mathcal{O}_{\sqrt{d}}$ is known to be a unique factorization domain:

$$d = \{2, 3, 5, 6, 7, 11, 13, 14, 17, 19, 21, 22, 23, 29, 31, 33, 37, 38, 41, 43, 46, 47, \dots\}$$

Example 6 (Euler). Every prime $p \equiv 1 \pmod{4}$ can be written as a sum of 2 squares.

Proof. By Wilson's theorem,

$$p \mid (p-1)! + 1 \equiv (-1)^{\frac{p-1}{2}} \left(\frac{p-1}{2}\right)^2 + 1 \equiv \left(\frac{p-1}{2}\right)^2 + 1 \pmod{p}.$$

i.e. $p \mid m^2 + 1$ for some integer m . Therefore, in $\mathbb{Z}[i]$ we have $p \mid m^2 + 1 = (z + i)(z - i)$. Since any $(a + bi)p$ has imaginary part an (integer) multiple of p , we cannot have $p \mid m \pm i$ in $\mathbb{Z}[i]$. Therefore, p is not prime in $\mathbb{Z}[i]$, but since $N(p) = p^2$ splits into 2 primes, p must decompose in the form $p = w\bar{w}$ for some $w \in \mathbb{Z}[i]$. Writing $w = a + bi$ we find $p = N(w) = a^2 + b^2$, as desired. $\square$

Example 7 (Lebesgue, 1850). For $n > 1$ the equation $x^2 + 1 = y^n$ has only the solution $x = 0, y = 1$ in integers.

Proof. If n is even then $x = 0, y = 1$, because $0, 1$ are the only consecutive perfect squares. We may assume $n = p$ is an odd prime. By mod 4 considerations x must be even and y is odd. Rewrite the equation as

$$(1 + ix)(1 - ix) = y^p$$

Let $d = \gcd(1 + ix, 1 - ix) \in \mathbb{Z}[i]$. Then $d \mid 1 + ix + 1 - ix = 2$ and $\bar{d} \mid \bar{2} = 2$ hence $d\bar{d} = N(d) \mid 4$. On the other hand, $d\bar{d} \mid (1 + ix)(1 - ix) = 1 + x^2$ which is odd. Therefore $N(d) = 1$, d is a unit and $1 + ix = a(u + iv)^p$ for some unit a . But p is odd, hence we may absorb a into the power and assume without loss $a = 1$. Now

$$1 + ix = (u + iv)^p \implies 1 = u^p - \binom{p}{2}u^{p-2}v^2 + \binom{p}{4}u^{p-4}v^4 - \dots \pm \binom{p}{p-1}uv^{p-1} \quad (1)$$

In particular $u = \pm 1$ and v is even. Taking (1) mod 4 we have $u^p \equiv 1 \pmod{4}$ hence $u = 1$ (as p is odd). Now (1) becomes

$$\binom{p}{2} = \binom{p}{4}v^2 - \binom{p}{6}v^4 + \dots \pm \binom{p}{p-1}v^{p-3} \quad (2)$$

Considering the exponent of 2 on both sides, note that

$$\binom{p}{2k+2}v^{2k} = \frac{p(p-1)\dots(p-2k-1)}{(2k)!}v^{2k}$$

For $k \geq 1$, we have $v_2((2k)!) < k + \frac{k}{2} + \frac{k}{4} + \dots = 2k \leq v_2(2k)$ and $v_2(p(p-1)\dots(p-2k-1)) > v_2(p(p-1))$, hence the exponent of 2 in any term on the right in (2) is larger than the exponent on the left, contradiction. The proof is complete. $\square$

Exercise. Solve example 5 by working in the *Eisenstein ring* $\mathbb{Z}[\omega]$ where $\omega^2 + \omega + 1 = 0$ (and $\omega^3 = 1$). Note that $a^2 - ac + c^2 = b^2 + bd + d^2 \iff (a - c\omega)(a - c\bar{\omega}) = (b + d\omega)(b + d\bar{\omega})$.

3 Working in extensions of $\mathbb{F}_p$

Example 8. Prove that $p \mid x^2 - 2 \implies p = 2$ or $p \equiv \pm 1 \pmod{8}$.

Solution. We know $p = 2$ works, so we assume p is odd. We have in $x^2 - 2 = 0$ in $\mathbb{F}_p$. We would like to construct an element for which it's easy to compute the order modulo p . Try $x = t + \frac{1}{t}$, i.e. take t to be a solution of the quadratic $t^2 - tx + 1 = 0$. We know $\mathbb{F}_p[t] \subseteq \mathbb{F}_{p^2}$. Now $(t + \frac{1}{t})^2 - 2 = 0 \implies t^4 = -1$ and $t^8 = 1$ so t has order 8 in $\mathbb{F}_p[t]$. If $t \in \mathbb{F}_p$ then $t^{p-1} = 1$ implies $8 \mid p-1$. Assume now $t \in \mathbb{F}_{p^2} - \mathbb{F}_p$. Since $x = t + \frac{1}{t}$ then $t^p + \frac{1}{t^p} = x^p = x = t + \frac{1}{t}$ and hence $\{t, \frac{1}{t}\} = \{t^p, \frac{1}{t^p}\}$ as the roots of the quadratic $t^2 - tx + 1 = 0$. Since $t \notin \mathbb{F}_p$, we must have $t = \frac{1}{t^p} \implies t^{p+1} = 1$. Therefore $8 \mid p+1$ in this case. $\square$

Example 9 (IMO Shortlist 2009). Let $a_0, a_1, a_2, \dots$ be a sequence defined by $a_0 = 2$ and $a_{k+1} = 2a_k^2 - 1, \forall k \geq 0$. Prove that if an odd prime p divides a_n then $p \equiv \pm 1 \pmod{2^{n+2}}$.

Proof. Using some observations and induction one can show that

$$a_n = \frac{(2 + \sqrt{3})^{2^n} + (2 - \sqrt{3})^{2^n}}{2}, \forall n \geq 0.$$

Suppose p is an odd prime dividing a_n . Because of $\sqrt{3}$ and remainders modulo p , it makes sense to work in $\mathbb{F}_p[\sqrt{3}]$ (this is either $\mathbb{F}_p$ or a quadratic extension of $\mathbb{F}_p$ – depending on whether 3 is a quadratic residue mod p). In this field, we have $(2 + \sqrt{3})(2 - \sqrt{3}) = 1$ hence

$$(2 + \sqrt{3})^{2^n} + (2 - \sqrt{3})^{2^n} = 0 \implies (2 + \sqrt{3})^{2^{n+1}} + 1 = 0 \quad (1)$$

However, note that, and $0 = a_n = 2a_{n-1}^2 - 1 \implies \frac{1}{2} = a_{n-1}^2$, hence

$$2(2 + \sqrt{3}) = (1 + \sqrt{3})^2 \implies (2 + \sqrt{3}) = ((1 + \sqrt{3})a_{n-1})^2$$

is a perfect square in $\mathbb{F}_p[\sqrt{3}]$. Now (1) implies

$$((1 + \sqrt{3})a_{n-1})^{2^{n+2}} = -1 \text{ and } ((1 + \sqrt{3})a_{n-1})^{2^{n+3}} = 1$$

which shows that the order of $(1 + \sqrt{3})a_{n-1}$ in $\mathbb{F}_p[\sqrt{3}]$ is 2^{n+3} . But this order must divide $|\mathbb{F}_p[\sqrt{3}]^*| \in \{p-1, p^2-1\}$, hence $2^{n+3} \mid p^2-1 = (p-1)(p+1) \implies p \equiv \pm 1 \pmod{2^{n+2}}$. $\square$

4 Practice Problems

Relations of type $ab = cd$

Problem 1. If a, b, c, d are positive integers such that $a^2 + b^2 = c^2 + d^2$, then

1. There exist positive integers $m, n, p, q, t \in \mathbb{N}^*$ such that

$$c = t(m^2 + n^2), d = t(p^2 + q^2), a = t(mp + nq), b = t(mq - np).$$

2. There exist positive integers m, n, p, q such that

$$2a = mn + pq, 2b = mp - nq, 2c = mp + nq, 2d = mn - pq.$$

Problem 2. Find all rational solutions of the system

$$\begin{cases} z^2 + r = u^2 \\ z^2 - r = v^2 \end{cases}$$

Problem 3. Find all integer solutions of

$$1. x^2 + 3y^2 = z^2 \quad 2. x^2 + y^2 = 5z^2$$

Problem 4. Show that if a, b, c, d, N are positive integers such that $N = a^2 + 2b^2 = c^2 + 2d^2$ and $(a, b) \neq (c, d)$, then N is composite.

Problem 5. Show that $2(x^4 - y^4) = z^2$ has no integer solutions.

Problem 6. Prove that there are no four (distinct) perfect squares forming an arithmetic progression.

Problem 7 (Vietnam 1998). Let d be a positive divisor of $5 + 1998^{1998}$. Show that d can be written in the form $d = 2x^2 + 2xy + 3y^2, x, y \in \mathbb{N}$ if and only if $d \equiv 3, 7 \pmod{20}$.

Working in quadratic extensions of $\mathbb{Q}$

Problem 8. Find all integer solutions to $y^2 + 2 = x^3$.

Problem 9. Prove that if $p \equiv 1 \pmod{6}$ then $p = a^2 - ab + b^2$ for some integers a, b .

Problem 10. Solve in positive integers $x^2 - xy + y^2 = z^2$.

Problem 11 (Balkan MO 1998). Solve in integers $x^2 + 4 = y^5$.

Problem 12 (Romania TST 1997). Let $A = \{a^2 + 2b^2 \mid a \in \mathbb{N}, b \in \mathbb{N}^*\}$. Show that if p is a prime such that $p^2 \in A$ then $p \in A$.

Problem 13. Suppose that x, y, z are positive integers such that $xy = z^2 + 1$. Then there exist integers a, b, c, d such that $x = a^2 + b^2, y = c^2 + d^2$ and $z = ac + bd$.

Problem 14 (Fermat). Prove that the equation $x^3 + y^3 = z^3$ has no non-trivial solutions in $\mathbb{Z}$.

Hint. Let $\rho = 1 - \omega$ in the Eisenstein ring $\mathbb{Z}[\omega]$. Then $1 - \rho$ is a prime of norm 3, for any $a \in \mathbb{Z}[\omega]$, exactly one of $a - 1, a, a + 1$ is divisible by ρ and the following lifting-the-exponent type property holds:

$$\alpha \equiv \pm 1 \pmod{\rho} \implies \alpha^3 \equiv \pm 1 \pmod{\rho^4}$$

Working in extensions of $\mathbb{F}_p$

Problem 15. Let $p \not\equiv \pm 1 \pmod{5}$ be a prime. Show that p divides the Fibonacci number F_{p+1} .

Problem 16. Let p, q be distinct primes such that q divides the Fibonacci number F_p . Show that $q \equiv \pm 1 \pmod{p}$.

Problem 17. Define the sequence (x_n) by $x_1 = 2, x_2 = 12$ and $x_{n+2} = 6x_{n+1} - x_n$. Let $q > 3$ be a prime factor of x_p , where p is prime. Show that $q \geq 2p - 1$.

Problem 18 (Brazil). Let p be a prime dividing $x^3 - 3x + 1$. Show that $p = 3$ or $p \equiv \pm 1 \pmod{9}$.